

MYAN GUPTA

(425) 900-4417 | gmyang03@gmail.com | <https://www.linkedin.com/in/myangupta/> | [Portfolio](#)

Education

University of Minnesota

Minneapolis, MN

Bachelor of Science (BS), Computer Science | GPA: 3.80 / 4.0

Sep 2023 – Dec 2026

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Databases, Artificial Intelligence, Machine Architecture, Modern Cryptography, Statistics, Computer Networks, Linear Algebra, Discrete Mathematics, Computer Graphics, Systems Programming, Cloud Computing, Machine Learning, Product Design and Development.

Technical Skills & Certifications

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Go, OCaml.

Backend & Cloud: FastAPI, Flask, REST APIs, Docker, Kubernetes, AWS (EC2, S3, Lambda), CI/CD, Linux/Unix, Azure.

AI/ML & Data: PyTorch, Hugging Face, OpenAI API, MLX, TPU Kernel Optimization, Pandas, NumPy, Claude, Cursor.

Tools & Core: Git, Playwright, Redis, PostGIS, Pydantic, GDB, Valgrind, ThreadSanitizer, k6, Load Testing, Concurrency, SQLite, Kuzu, LanceDB, Typer, Pytest, VectorDB, RAG.

Work Experience

University Of Minnesota, Research Assistant

Jan 2025 – Present

- Built trace-analysis workflows for multi-agent LLM systems using Python and FastAPI, categorizing system design, inter-agent coordination, and task-verification failures across execution runs, which helped the team pinpoint design flaws more quickly.
- Developed scalable evaluation pipelines inspired by MAST-Data to analyze 1,600+ annotated traces across 7 agent frameworks and 14 failure modes, improving root-cause diagnosis for unreliable agent behavior.

University Of Minnesota, Teaching Assistant

Sep 2024 – Present

- Mentored and led labs for 100+ students in object-oriented programming, algorithms, debugging, and systems concepts, improving exam performance by 20%.
- Helped students resolve code defects in assignments and designed algorithm labs, raising average lab scores by 25% while improving submission quality.

Projects

Kernel Factory ([Link](#))

- Architected a deterministic TPU kernel-generation pipeline that converts model layer specs into verified kernels for GPT-2-style workloads, making low-level JAX Pallas optimization accessible.
- Designed a VMEM-aware TileSolver, template retrieval system, assembler, and compiler verification gate to compute hardware-valid tile configurations and prevent unsafe free-form kernel code generation.
- Generated and numerically verified 13 TPU kernels on Google Cloud TPU v5e, achieving 1.00x XLA parity on MatMul, 1.11x faster fused MatMul+RMSNorm, 5% faster RMSNorm, and 92 passing tests.

Reinforcement Learning Benchmarking Platform ([Link](#))

- Created an OpenEnv-compatible Reinforcement Learning evaluation backend for email workflows, achieving 0.967 mean return across benchmark tiers.
- Developed an async JSON-RPC MCP server with Pydantic validation to run 20+ classification, drafting, and workflow scenarios through a standardized API.
- Added deterministic grading, trace logging, and integration tests across 3 benchmark tiers, improving reproducibility for multi-step agent tasks.

Consistent Hashing Key-Value Store ([Link](#))

- Engineered a distributed key-value store with hash-ring routing for 1–3 backend nodes to reduce key remapping.
- Added CRUD, health-check, ring-inspection, and node-debug endpoints to inspect routing and service state.
- Benchmarked 155 requests/sec with 136ms P95 latency and 0% failure rate under multiple node configurations.

ICU Capacity Forecasting Platform ([Link](#))

- Deployed a containerized ML forecasting platform on AWS EC2, S3, and Lambda with REST APIs and a React dashboard to support hospital capacity planning across 8 hospital clients.
- Reduced P95 forecast-retrieval latency from 360ms to 14ms by optimizing API queries and concurrency bottlenecks under load.
- Integrated SARIMA, Prophet, and federated learning, improving model robustness on non-IID healthcare time-series data.